BTECHBUDDY - Data Structures & Algorithms

Unit 1 to 5 Comprehensive Topper Handwritten Notes

Subject: CS201 | Branch: CSE / IT / AIML | 3rd Semester

Topics Covered:

1. Asymptotic Notations: Big-O, Big-Omega, Big-Theta Definitions
2. Linear Data Structures: Arrays, Singly/Doubly/Circular Linked Lists
3. Stacks & Queues: Array implementation, Infix to Postfix conversion
4. Trees: Binary Tree traversals Inorder, Preorder, Postorder, Level-order
5. Balanced Trees: AVL Rotations LL, RR, LR, RL with balance factor
6. Graphs: BFS, DFS, Dijkstra Shortest Path, Prim and Kruskal MST
7. Dynamic Programming: 0/1 Knapsack, Longest Common Subsequence LCS

Key Exam Tips: Always state the base condition in recursive routines.

BTECHBUDDY verified study resource. Visit https://btechbuddy.in for video lectures.